Unit: 6 Unit 9: Data Statistical Analysis **Name:** ______________
Topic: Measures of Central Tendency **Date:** __________

Advanced (Habanero)

Q1. What is the mean of $2, 4, 6, 8, 10$?
(A) 4
(B) 5
(C) 6
(D) 8
(E) 10

Q2. What is the median of $1, 3, 5, 7, 9$?
(A) 3
(B) 5
(C) 7
(D) 9
(E) 11

Q3. What is the mode of $2, 2, 2, 4, 4$?
(A) 2
(B) 4
(C) 6
(D) 8
(E) No mode

Q4. A dataset has a mean of 10 and a median of 12. What can be said about the dataset?
(A) It is symmetric
(B) It is skewed to the right
(C) It is skewed to the left
(D) It has an outlier
(E) Nothing can be said

Q5. Two datasets have the same median but different means. What can be said about the datasets?
(A) They have the same range
(B) They have the same mode
(C) They have different standard deviations
(D) They have different variances
(E) Nothing can be said

Q6. A dataset has a mean of 15 and a standard deviation of 3. What is the z-score of 12?
(A) -1
(B) 0
(C) 1
(D) 2
(E) 3

Q7. What is the purpose of calculating the mean, median, and mode?
(A) To determine the range of a dataset
(B) To compare datasets
(C) To determine the central tendency of a dataset
(D) To calculate the standard deviation
(E) To create a graph

Q8. Which measure of central tendency is most affected by outliers?
(A) Mean
(B) Median
(C) Mode
(D) Range
(E) Standard deviation

Open Ended Questions (FRQ)

Q9. Calculate the mean, median, and mode of the dataset $2, 4, 6, 8, 10$. Show your work and explain your answers.

Q10. Compare and contrast the mean, median, and mode. How are they used in real-world applications? Provide examples.

Chef's Tips

- Tip 1: Remind students to show their work for FRQs
- Tip 2: Encourage students to use calculators for calculations
- Tip 3: Emphasize the importance of labeling answers clearly